CGU ELE

Digital Integrated Circuit Design
HW-2: Memory Read/Write Control

Problem:

Please design read/write control to access memory based on the following specifications.

1.1 A register file has 32 words with 24-bit width. The module name and the input/output signals are listed as follows,

```
module RF1_32x24b(Q,     // data output bus
                  CLK,   // clock
                  CEN,   // chip enable
                  WEN,   // write active low, read active high
                  A,     // address bus
                  D);    // data input bus
```

1.2 The block diagram of the memory read/write access control is shown in Fig. 2. All of signals, **Data_In** (the data input of memory), **Data_Out** (the data output of memory), **WENAB** (the write enable with active "Lo"), **RENAB** (the read enable with active "Hi"), **CLK**, and **RESET_**, are provided in testbench, **HW2_WRMEM_rtb.v**. In order to show the end for both write and read operations, your design has to generate **WRMEM_Done** (Write operation done) and **RDMEM_Done** (Read operation done), respectively. Both **WRMEM_Done** and **RDMEM_Done** indicate that the **write** and the **read** operations are completely finished, respectively.

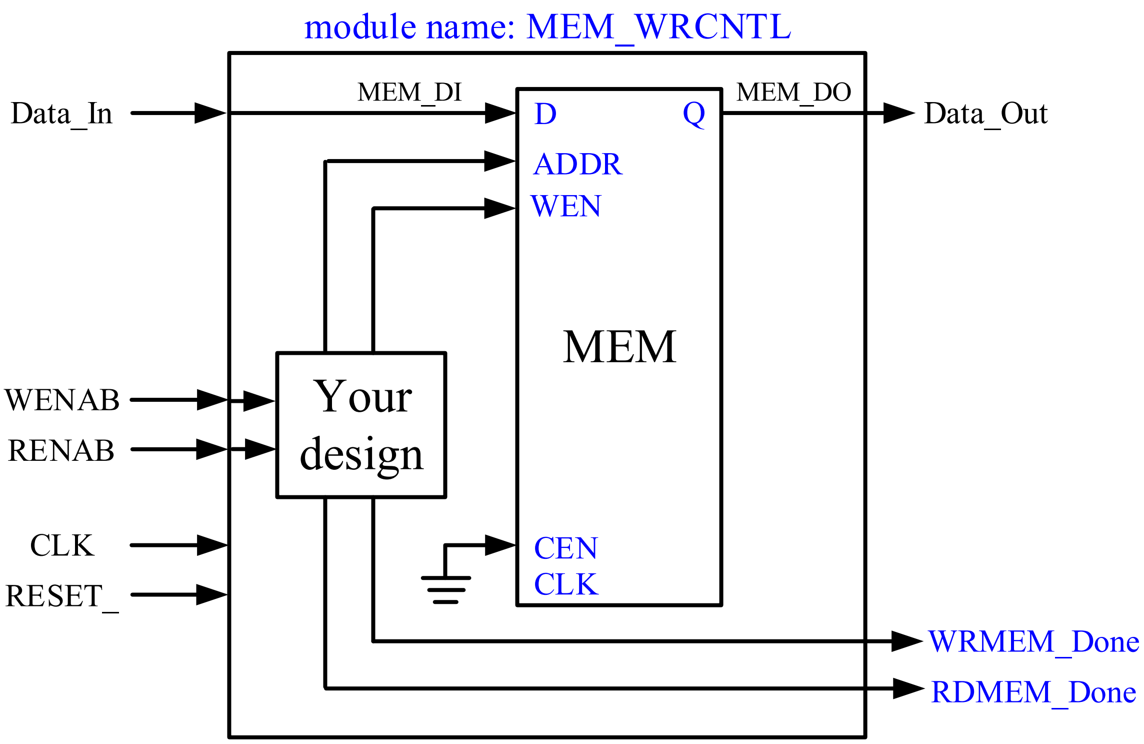


Fig. 1. Block diagram of the memory read/write access control

1.3 The template Verilog RTL **HW2_WRMEM_rtl.v** is also uploaded in E-learning, where your design should be added in **HW2_WRMEM_rtl.v**.

1.4 All of the **registers** in your design are assumed to operate in the **rising** edge-triggered.

1.5 In order to emulate the hardware behavior, please add a delay time to register in your Verilog RTL. The delay can truly emulate the delay time of the clock-to-Q for a register. An example of counter is described as follows,

Ex.

```
#define DLY #1
:
always @(posedge CLK or negedge RESET_)
   if (!RESET_)
      CNT <= `DLY 0;
   else
      CNT <= `DLY CNT+1;
:
```

1.6 The related write timing wave is shown in Fig. 2.

1.7 The related read timing wave is illustrated in Fig. 3.

1.8 Please design a read/write control to access RF1_32x25b (**RF1_32x25b.v**) based on a given testbench. Of course, you can try to modify the testbench to evaluate your RTL code.

Submission

1. Please submit the following files to E-learning.

(a) B123456_HW2_rtl.v: Your top-module design, where **B123456** is your **student ID**. If you have another RTL code for decoder, for instance, please name as B123456_DECODER.v.

(b) B123456_HW2_rtb.v: A testbench for you top-module design

2. Please deliver the simulation results and the circuit diagrams of your design (.docx) in class.

(a) For the simulation results, please attach the related photos (.jpg or pdf) in .docx.

(b) For the circuit diagrams, please sketch it using VISIO (.vsdx) or pen and then scan it (.jpg or .pdf). After that, please paste the circuit diagram in .docx.

Fig. 2. Write waveform

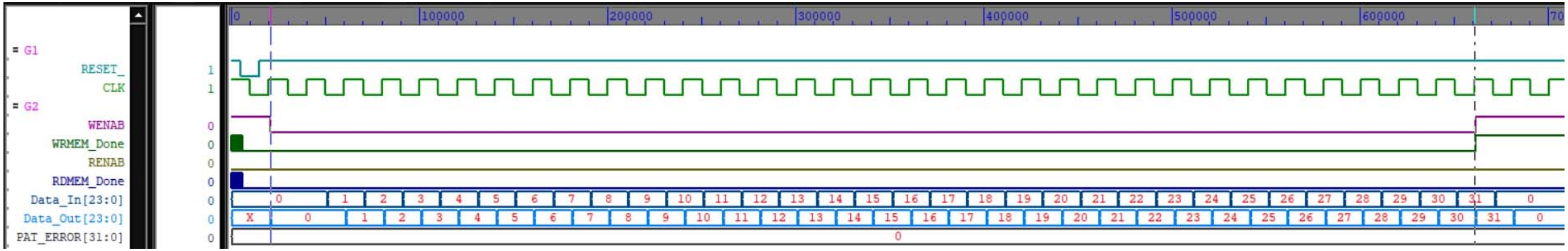
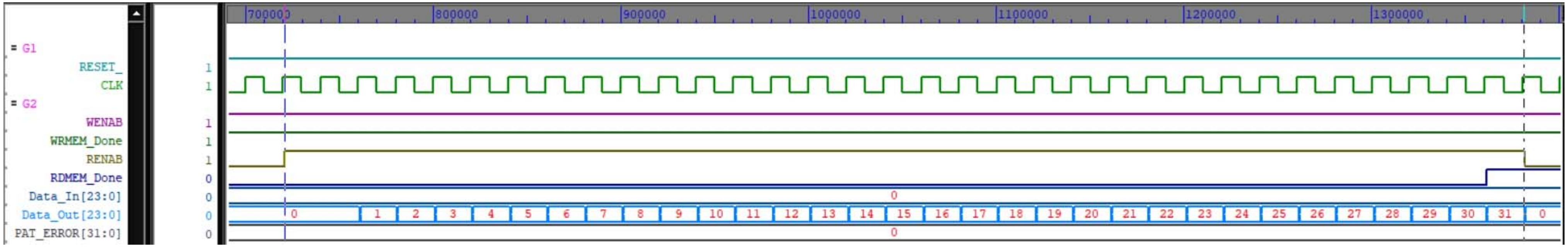


Fig. 3. Read waveform



Signal description for Fig. 4 and 5

(1) ADDR_CNT: An address counter is used to generate the access address for write and read operations.

(2) ACNT_ENAB: An enable signal to trigger the operation of the address counter for performing plus one operation

Fig. 4 Write waveform with the related control signals

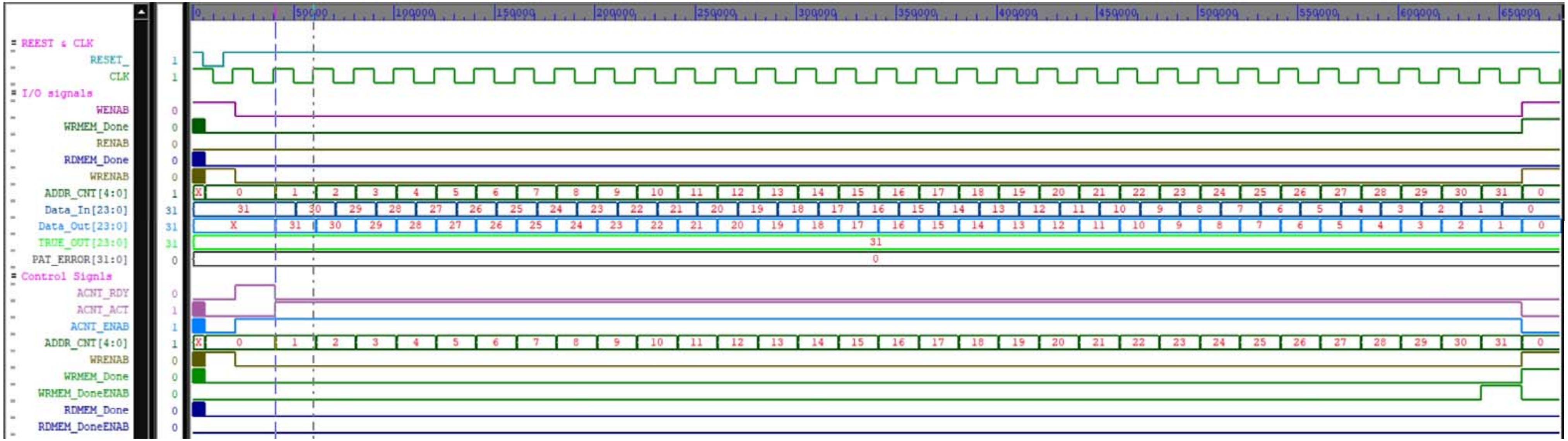


Fig. 5 Read waveform with the related control signals